EXPERIMENT : 15

Objective : Single Inheritance

Source Code

```
#include <iostream>
using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_pc() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_pc() {
        cout << "\nID = " << id << endl;
        cout << "Name = " << name << endl;
    }
};

class Student : public Person {
public:
```

```
    int fee;
    char course[100];

    void set_sc() {
        cout << "Enter Fee: ";
        cin >> fee;
        cout << "Enter Course: ";
        cin >> course;
    }

    void display_sc() {
        cout << "Fee = " << fee << endl;
        cout << "Course = " << course << endl;
    }
};

int main() {
    Student obj;

    obj.set_pc();
    obj.set_sc();

    obj.display_pc();
    obj.display_sc();

    return 0;
```

}

Output:

```
C:\Users\HP\OneDrive\Docun

Enter ID: 17521
Enter Name: Abhishek.Sharma
Enter Fee: 135000
Enter Course: CSE

ID = 17521
Name = Abhishek.Sharma
Fee = 135000
Course = CSE

--------------------------------
Process exited after 43.42 seconds with return value 0
Press any key to continue . . .
```

<u>**EXPERIMENT : 16**</u>

Objective : Multiple Inheritance

Source Code:

```
#include <iostream>
using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_pc() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_pc() {
        cout << "ID = " << id << endl;
        cout << "Name = " << name << endl;
    }
};

class College {
```

```
public:
    char college_name[100];

    void set_college() {
        cout << "Enter College Name: ";
        cin >> college_name;
    }

    void display_college() {
        cout << "College = " << college_name << endl;
    }
};

class Student : public Person, public College {
public:
    void show() {
        display_pc();
        display_college();
    }
};

int main() {
    Student obj;
    obj.set_pc();
    obj.set_college();
    obj.show();
```

```
    return 0;

}
```

<u>Output:</u>

```
C:\Users\HP\OneDrive\Docun

Enter ID: 17521
Enter Name: Abhishek.Sharma
Enter College Name: SVVV
ID = 17521
Name = Abhishek.Sharma
College = SVVV

--------------------------------
Process exited after 22.39 seconds with return value 0
Press any key to continue . . .
```

EXPERIMENT : 17

Objective: Hierarchical Inheritance

Source Code :

```
#include <iostream>
using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_person() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_person() {
        cout << "ID = " << id << endl;
        cout << "Name = " << name << endl;
    }
```

```
};

class Student : public Person {
public:
    void display_student() {
        cout << "This is Student class.\n";
    }
};

class Teacher : public Person {
public:
    void display_teacher() {
        cout << "This is Teacher class.\n";
    }
};

int main() {
    Student s;
    Teacher t;

    cout << "Student Info:\n";
    s.set_person();
    s.display_person();
    s.display_student();

    cout << "\nTeacher Info:\n";
```

```
    t.set_person();

    t.display_person();

    t.display_teacher();


    return 0;

}
```

Output:

```
C:\Users\HP\OneDrive\Docun
Student Info:
Enter ID: 17521
Enter Name: Abhishek.Sharma
ID = 17521
Name = Abhishek.Sharma
This is Student class.

Teacher Info:
Enter ID: 52100
Enter Name: Devendra.kuril.sir
ID = 52100
Name = Devendra.kuril.sir
This is Teacher class.

--------------------------------
Process exited after 40.49 seconds with return value 0
Press any key to continue . . .
```

EXPERIMENT : 18

Objective : Hybrid Inheritance

Source Code :

```
#include <iostream>
using namespace std;

class Person {
public:
   int id;
   char name[100];

   void set_person() {
      cout << "Enter ID: ";
      cin >> id;
      cout << "Enter Name: ";
      cin >> name;
   }

   void display_person() {
      cout << "ID = " << id << endl;
      cout << "Name = " << name << endl;
   }
```

```
};

class College {
public:
    char cname[100];

    void set_college() {
        cout << "Enter College Name: ";
        cin >> cname;
    }

    void display_college() {
        cout << "College = " << cname << endl;
    }
};

class Student : public Person {
public:
    char course[100];

    void set_student() {
        cout << "Enter Course: ";
        cin >> course;
    }

    void display_student() {
```

```cpp
        cout << "Course = " << course << endl;
    }
};

class Result : public Student, public College {
public:
    void show_all() {
        display_person();
        display_student();
        display_college();
    }
};

int main() {
    Result r;

    r.set_person();
    r.set_student();
    r.set_college();

    r.show_all();

    return 0;
}
```

Output:

```
C:\Users\HP\OneDrive\Docun

Enter ID: 17521
Enter Name: Abhishek.Sharma
Enter Course: CSE
Enter College Name: SVVV
ID = 17521
Name = Abhishek.Sharma
Course = CSE
College = SVVV

--------------------------------
Process exited after 23.71 seconds with return value 0
Press any key to continue . . .
```

EXPERIMENT : 19

Objective : Write a function with no argument and no return value

Source

```
#include <iostream>
using namespace std;
void arithmatic()
{
   float a,b;
   cout<<"enter a";
   cin>>a;
   cout<<"enter b";
   cin>>b;

   cout<<"the sum of a and b is = "<<a+b<<endl;
   cout<<"the multiplication of a and b is = "<<a*b<<endl;
   cout<<"the subtraction of a and b is = "<<a-b<<endl;
   cout<<"the division of a and b is = "<<a/b<<endl;
}
int main()
{
   arithmatic();
}
```

OUTPUT:

```
C:\Users\HP\OneDrive\Docun
enter a50
enter b50
the sum of a and b is = 100
the multiplication of a and b is = 2500
the subtraction of a and b is = 0
the division of a and b is = 1

--------------------------------
Process exited after 31.35 seconds with return value 0
Press any key to continue . . .
```

EXPERIMENT : 20

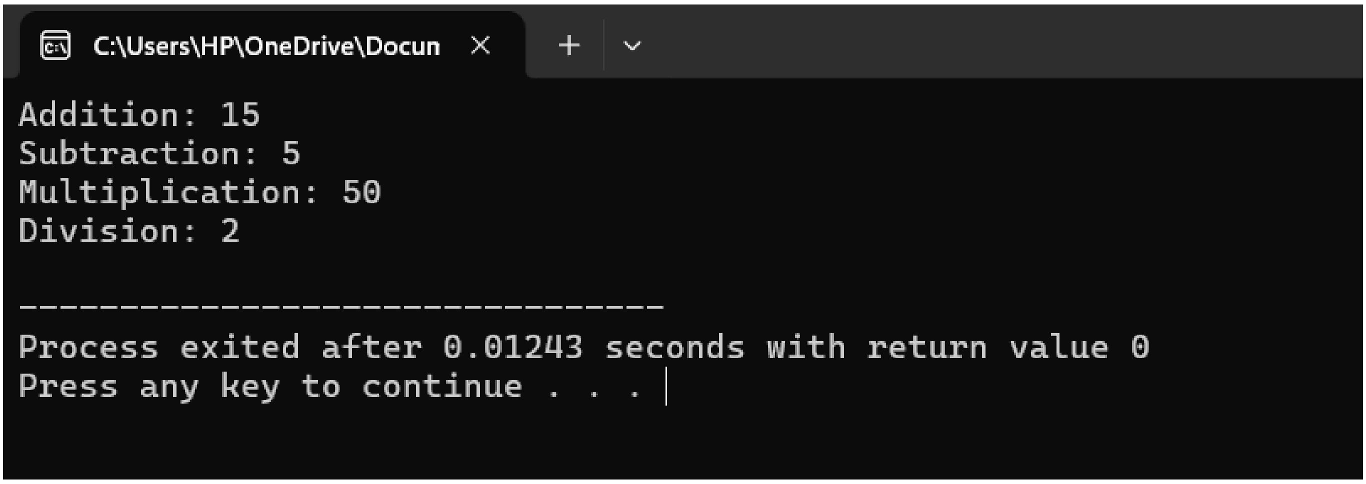
Objective : Write a function with no argument & return value.

Source Code

```
#include <iostream>
using namespace std;

int add() {
    int a = 100, b = 50;
    return a + b;
}
int subtract() {
    int a = 100, b = 50;
    return a - b;
}
int multiply() {
    int a = 100, b = 50;
    return a * b;
}
float divide() {
    float a = 100, b = 50;
    return a / b;
}

int main() {
    cout << "Addition: " << add() << endl;
```

```
    cout << "Subtraction: " << subtract() << endl;

    cout << "Multiplication: " << multiply() << endl;

    cout << "Division: " << divide() << endl;


    return 0;

}
```

OUTPUT :



EXPERIMENT : 21

Objective : Write the function with argument &no return value

Source:

```
#include <iostream>
using namespace std;
void arithmatic(int a , int b){

   int sum = a+b;
   int subtraction = a-b;
   int multiplication= a*b;
   int  division = a/b;
   cout<<"the sum is ="<<sum<<endl;
   cout<<"the division is ="<<division;
   cout<<endl;
   cout<<"the multiplication is ="<<multiplication;
   cout<<endl;
   cout<<"the subtraction is ="<<subtraction;
}
int main() {
   arithmatic(100 , 25);
   return 0;

}
```

Output:

```
C:\Users\HP\OneDrive\Docun

the sum is =125
the division is =4
the multiplication is =2500
the subtraction is =75
--------------------------------
Process exited after 0.008649 seconds with return value 0
Press any key to continue . . .
```

EXPERIMENT : 22

Objective: write a function with argument & return value

Source Code :

```
#include <iostream>
using namespace std;
int add(int a , int b) {
   return a + b;
}
int subtract(int a , int b) {
    return a - b;
}
int multiply(int a , int b) {
    return a * b;
}
float divide(float a , float b) {
  return a / b;
}
int main() {
   cout << "Addition: " << add(100 , 30) << endl;
   cout << "Subtraction: " << subtract(50,60) << endl;
   cout << "Multiplication: " << multiply(6,7) << endl;
   cout << "Division: " << divide(10,9) << endl;
 return 0;
}
```

<u>Output:</u>

```
C:\Users\HP\OneDrive\Docun

Addition: 130
Subtraction: -1
Multiplication: 72
Division: 0.888889

--------------------------------
Process exited after 0.01099 seconds with return value 0
Press any key to continue . . .
```